01 Execution Evidence Report

Step-by-step build evidence, screenshots, validation, decisions, and ownership disclosure.

Repository	https://github.com/aalanani2000/al-rouf-ai-integration-assessment
Default branch	main
Latest commit	653ec552871083dca14498d120d6e6c156ed4f31
Prepared date	2026-06-24
Assessment	AI Integration Engineer task pack

Assessment Requirements Covered

- Runnable GitHub repository with setup documentation, task folders, screenshots, and report deliverables.
- RFQ to CRM automation using n8n as an equivalent orchestration approach.
- FastAPI quotation microservice with tests, Docker packaging, and OpenAPI documentation.
- Bilingual RAG workflow over English and Arabic sample documents with citations and refusal behavior.
- Explicit notes on error handling, maintainability, security hygiene, trade-offs, validation, and AI assistance.

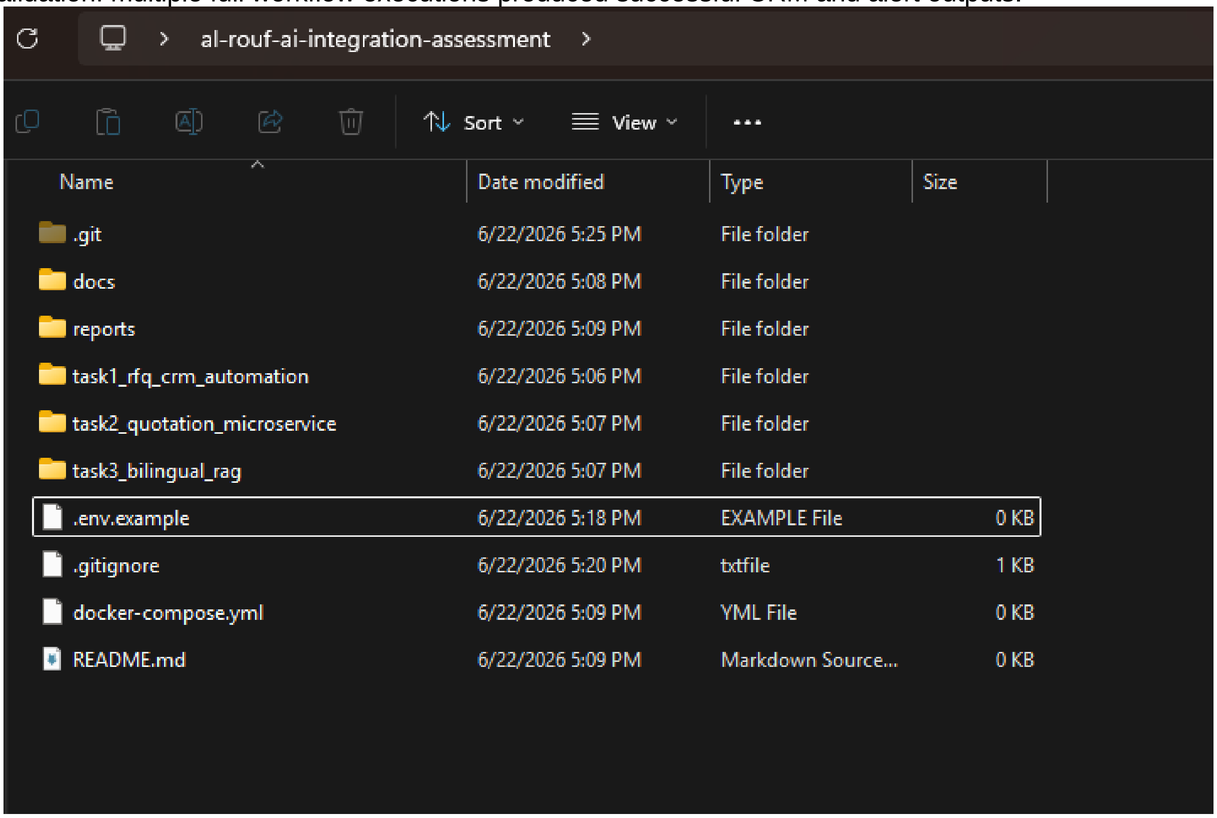
Work Evidence By Task

Task	Implementation	Validation	Status
Task 1	n8n RFQ webhook, OpenAI extraction, mock CRM, bilingual reply, alert log.	Successful n8n executions, persisted JSON records, screenshot evidence.	Complete; attachment bytes scoped as future enhancement.
Task 2	FastAPI quotation API with catalog, pricing tiers, SQLite, Docker, OpenAPI.	Automated pytest suite and OpenAPI screenshot evidence.	Complete.
Task 3	FastAPI bilingual RAG service over English/Arabic documents with citations and refusal.	Unit tests, Arabic question evidence, error-handling screenshot.	Complete.

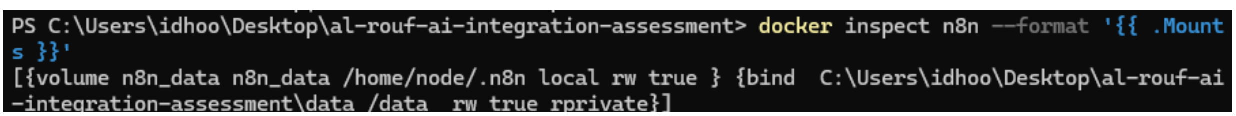
Task 1 Evidence: RFQ To CRM Automation

1. Created a Docker-based n8n environment with persistent volumes.
2. Built a webhook intake endpoint for RFQ-style JSON payloads.
3. Configured OpenAI extraction for urgency, key requirements, and summary.
4. Merged structured fields with extracted intent and generated CRM metadata.
5. Persisted CRM records and sales alerts with bilingual reply drafts.
6. Debugged sandbox, missing-file, and node-reference issues during testing.

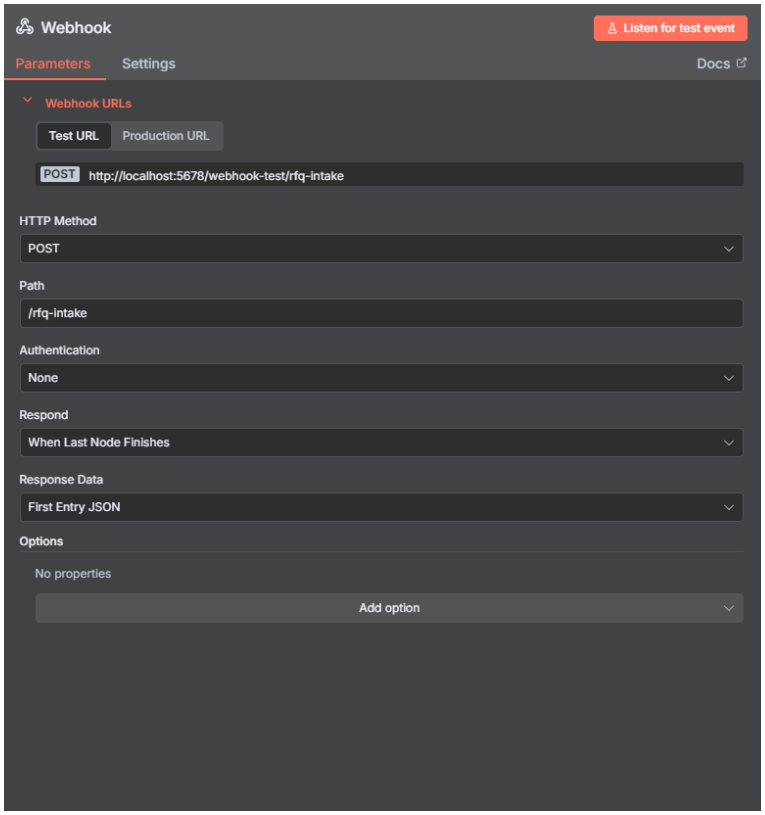
- Trade-off: webhook simulation kept the workflow reproducible without OAuth setup.
- Limitation: attachment filenames are captured as metadata; real file storage is a future enhancement.
- Validation: multiple full workflow executions produced successful CRM and alert outputs.



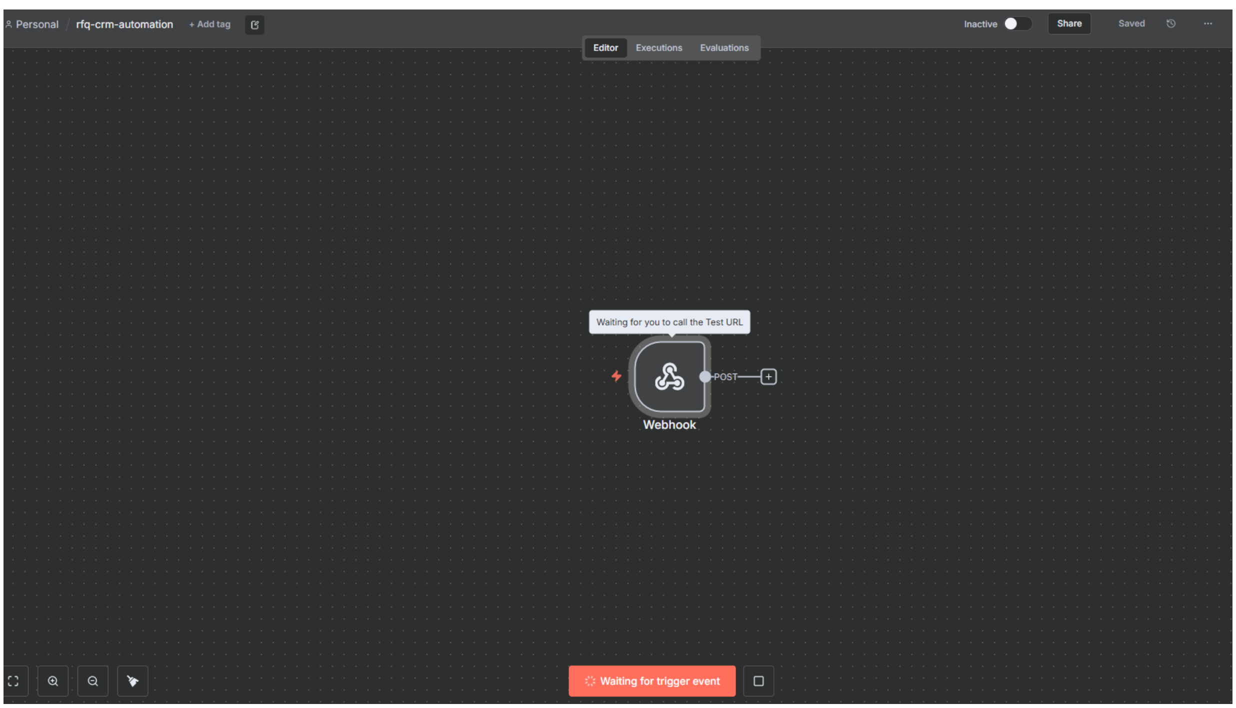
Task 1 evidence: repository structure.png



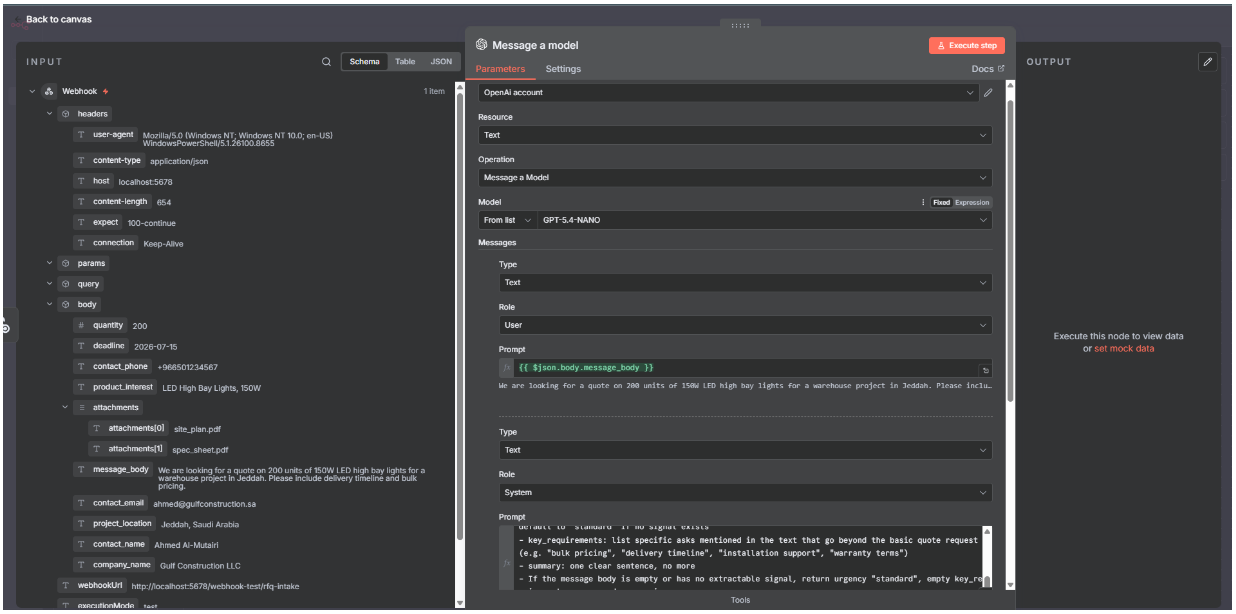
Task 1 evidence: proof of the volumes being correctly attached.png



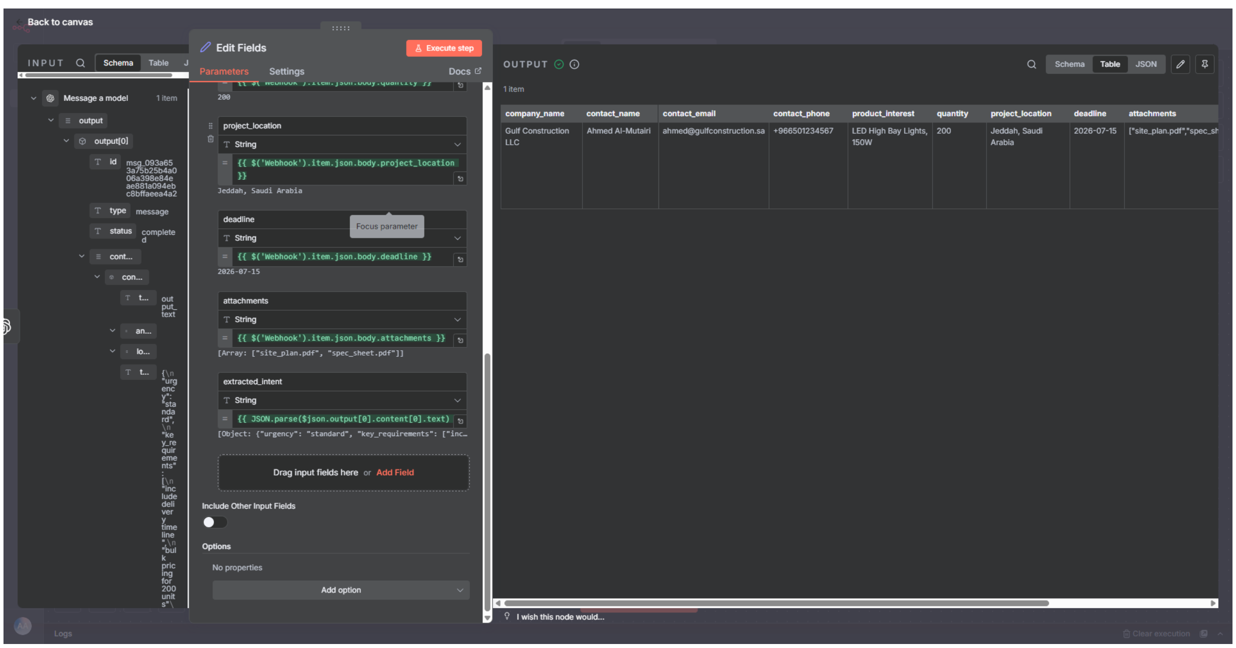
Task 1 evidence: Webhook node.png



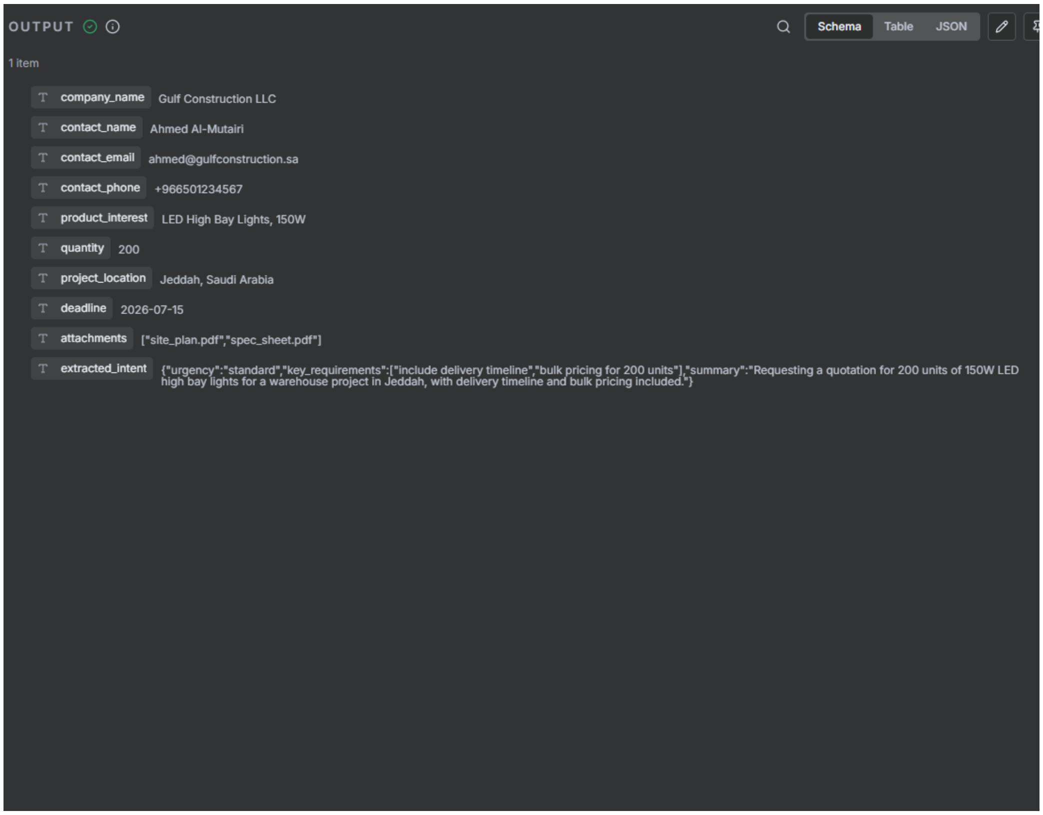
Task 1 evidence: n8n canva.png



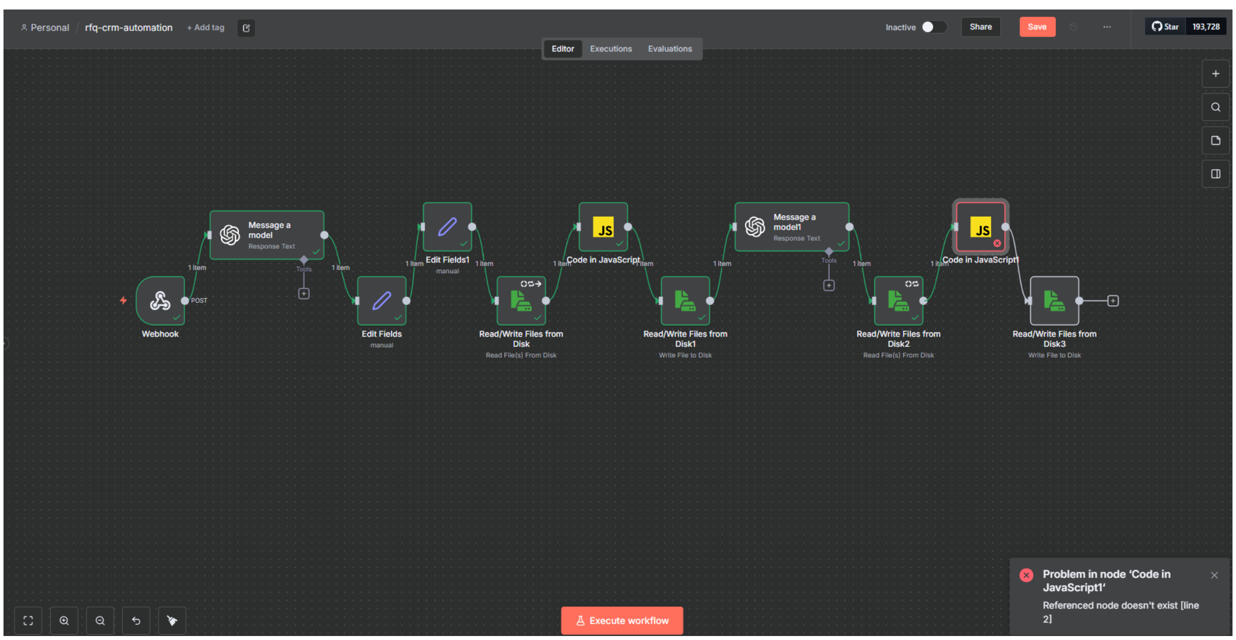
Task 1 evidence: open ai node.png



Task 1 evidence: set field.png

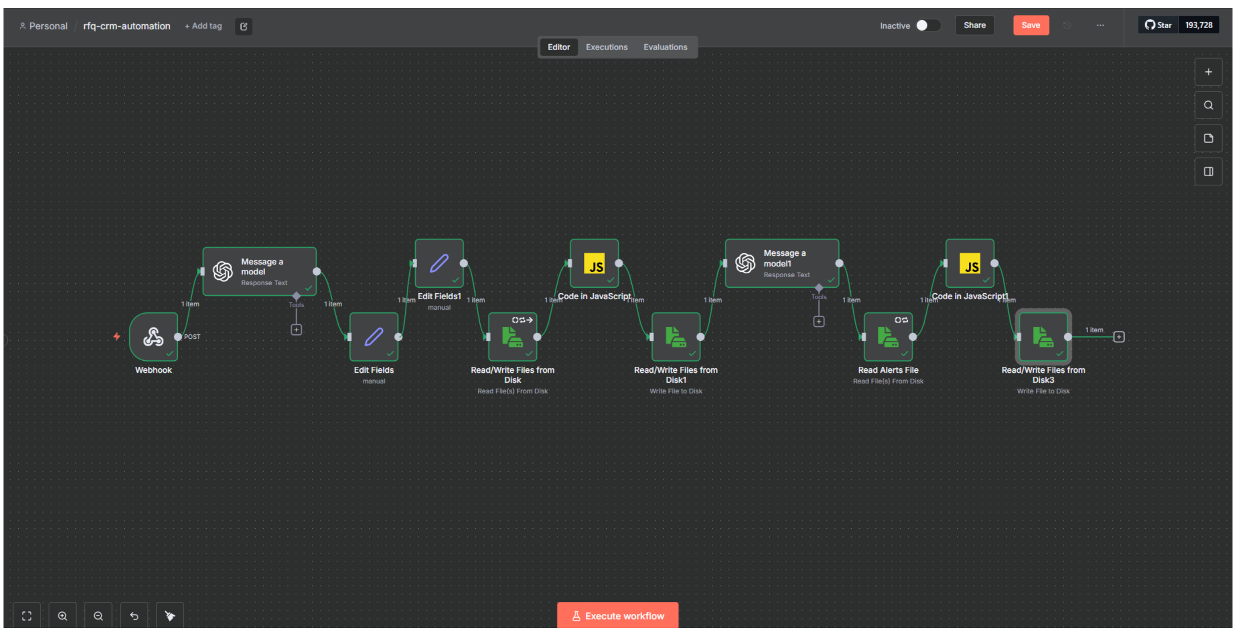


Task 1 evidence: set field output.png

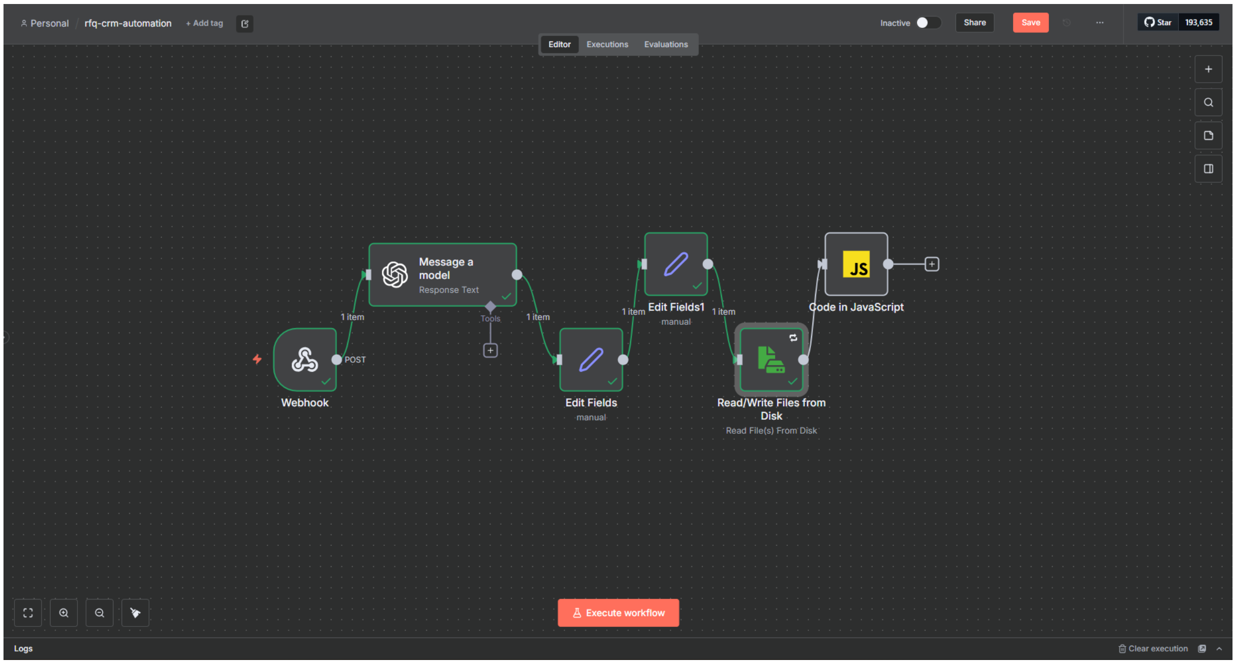


Task 1 evidence: javascript code 1 error .png

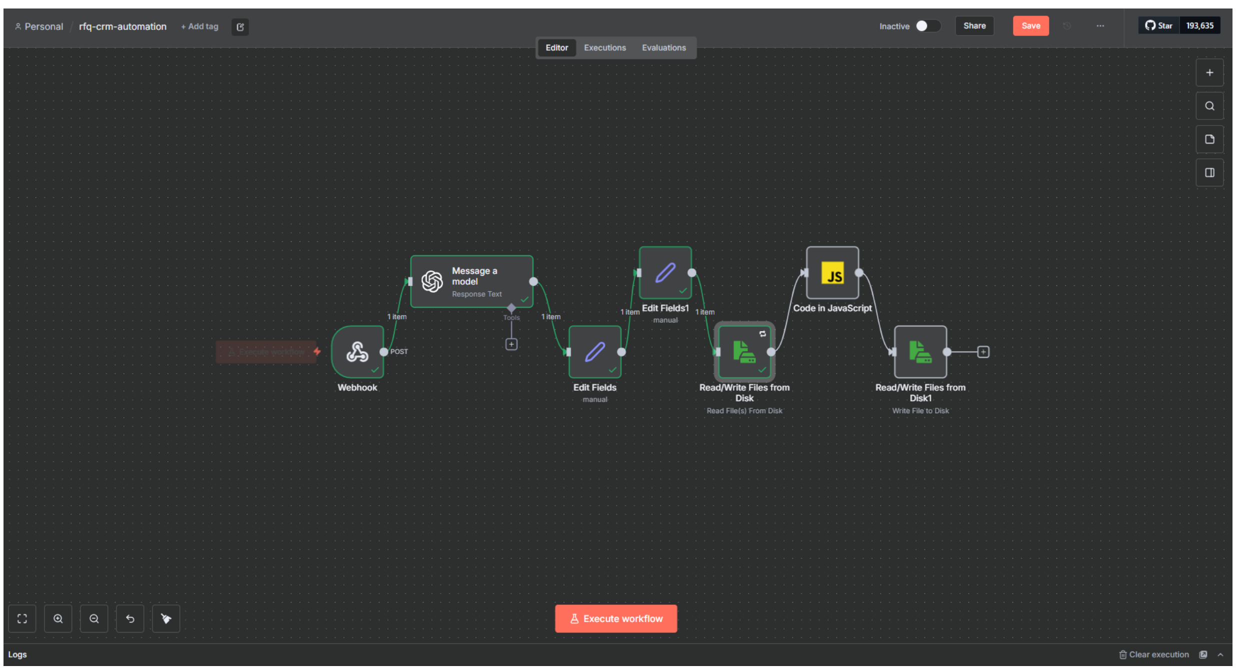
Task 1 evidence: workflow progress .png



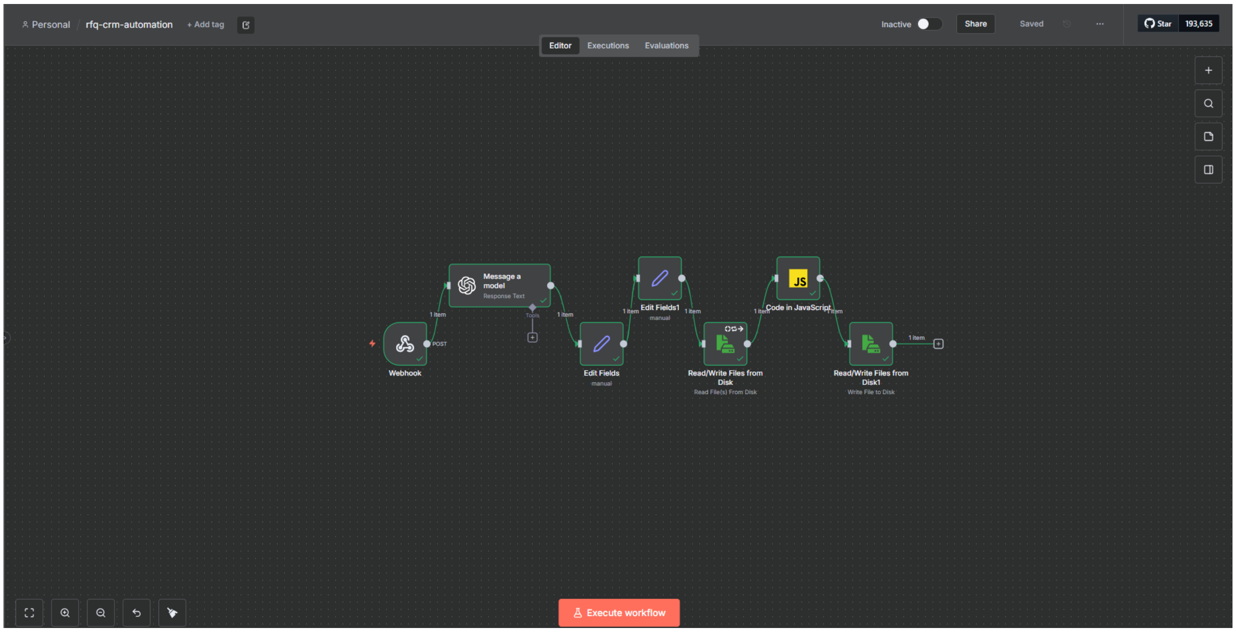
Task 1 evidence: workflow progress 1 .png



Task 1 evidence: workflow.png



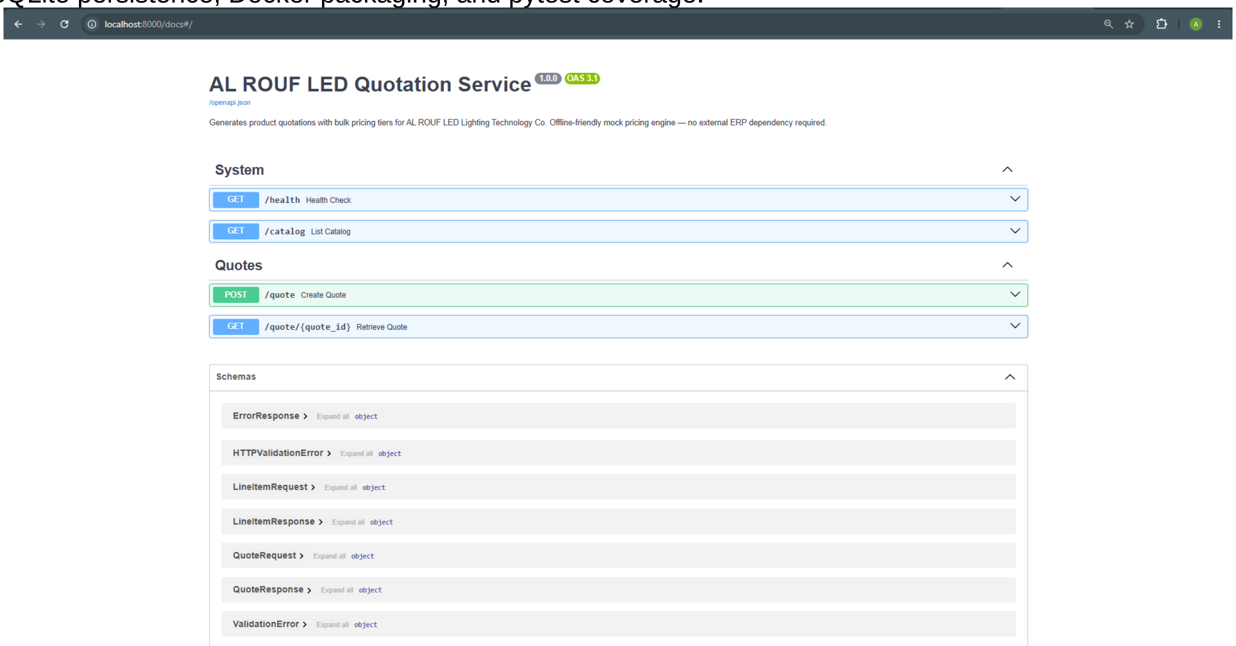
Task 1 evidence: workflow2.png



Task 1 evidence: workflow succes.png

Task 2 Evidence: Quotation Microservice

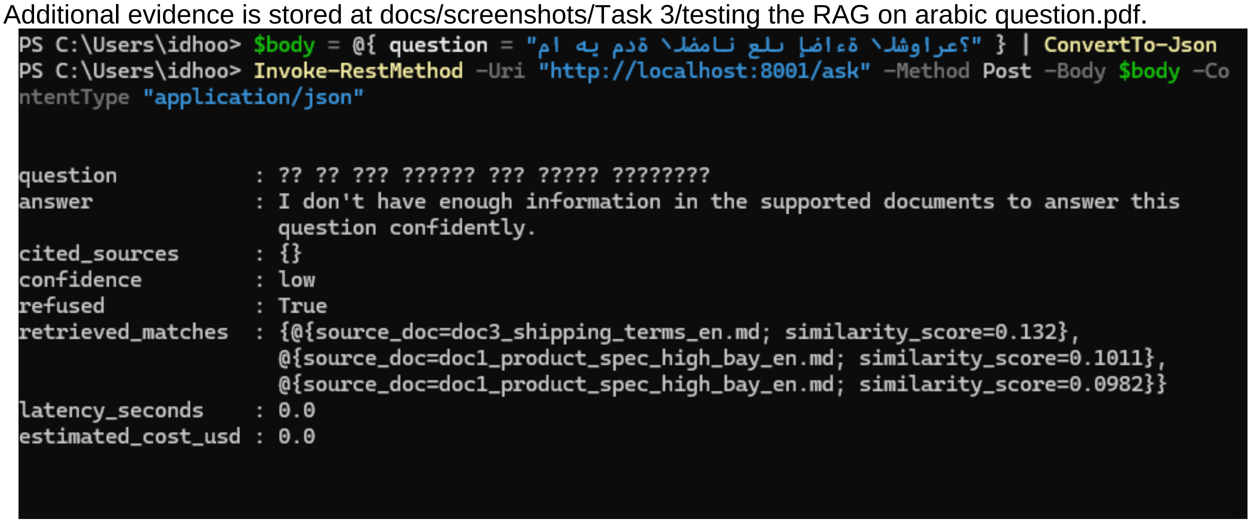
- FastAPI service with health, catalog, quote creation, and quote retrieval endpoints.
- Product catalog and bulk-discount logic with offline-friendly mocked data.
- SQLite persistence, Docker packaging, and pytest coverage.



Task 2 evidence: OpenAPI documentation page.

Task 3 Evidence: Bilingual RAG Knowledge Workflow

- FastAPI service loads 4 sample documents in English and Arabic.
- Embeddings are cached locally and searched with cosine similarity.
- The service returns cited answers and refuses out-of-scope questions.
- Additional evidence is stored at docs/screenshots/Task 3/testing the RAG on arabic question.pdf.



Task 3 evidence: model communication issue captured during testing.

Validation Summary

- Task 1 was validated through end-to-end n8n executions and persisted mock output files.
- Task 2 includes a pytest suite and OpenAPI evidence.
- Task 3 includes offline tests plus optional API-key integration tests.
- Docker packaging is included for the runtime services.

Ownership And AI Assistance Disclosure

AI assistance was used as a technical mentor and documentation partner for architecture discussion, starter logic, debugging support, and final writing polish. The candidate personally completed the hands-on implementation work: n8n node setup, Docker execution, API implementation, test execution, screenshot collection, environment configuration, validation, and final engineering decisions.